Master’s Degree in Data Science
Academic Year: 2025-2026

LAB PROJECT

MARKETING MIX MODELLING

PRESENTED BY : Matteo Fumagalli - id. 894468

MARKETING ANALYTICS

EXECUTIVE SUMMARY

Objective

The goal of this project is to develop and interpret a Bayesian **Marketing Mix Model (MMM)** to maximize total revenue and optimize budget allocation across multiple media channels.

Business Scenario

Acting as a marketing data scientist for a multi-channel e-commerce company, this analysis addresses specific challenges expressed by senior management, such as:

- Determining the individual **contribution of each marketing channel** to overall purchases.
- Estimating both **channel-level ROI and mROI** to identify efficiency benchmarks.
- Building **future investment scenarios** to predict the business impact of budget reallocations.

Key Findings about ROI

Meta Facebook currently receives the largest share of the marketing budget and drives the majority of total revenue. However, this channel is approaching its saturation point, with an ROI near 1.0. In contrast, Google Search-based channels show higher ROIs and capture active consumer behaviour rather than passive exposure.

Recommended Spend Allocation

To maximize total revenue, it is necessary to shift the focus from high-spend, saturated channels towards under-invested platforms that have not yet reached their saturation threshold. By doing this, the organisation can improve its overall efficiency and optimize the total return on advertising investment without necessarily increasing the available budget.

METHODOLOGY

DATASET PREPARATION AND EXPLORATION

Dataset Overview

The original dataset includes **daily observations** from 21 July 2021 to 2 June 2024, covering 93 organisations across various sectors and territories. To ensure the analysis is meaningful and reliable, data for a single organisation was extracted according to specific criteria: it operates in multiple territories, it does not contain missing days or duplicates, and it maintains at least one channel with spending exceeding 40% of the total period.

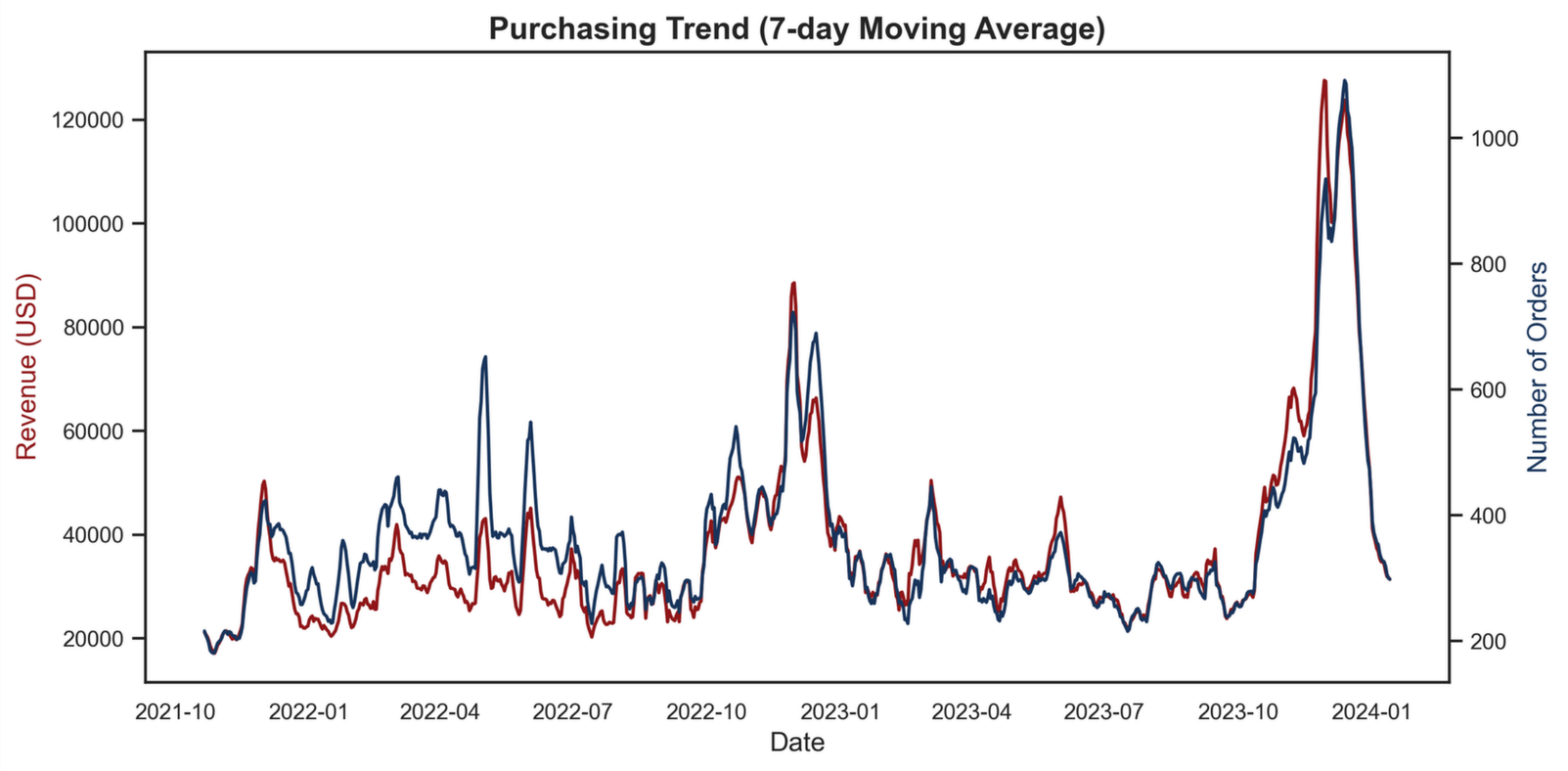
The **selected organisation** (code 7569a6a9c156a0f9398fa6cfd51df5bb) operates in the Food & Drink sector across the United States, Canada, and the United Kingdom. Specifically, the analysis focuses on the **“All Territories” subset** to evaluate the organisation’s performance at a global level.

Missing Values

Data quality assessment confirms that there are no missing values in core variables, such as date, revenue, or purchase counts. However, **some missing values are identified within the media columns**, specifically for spend, clicks, and impressions. These instances are imputed to zero, assuming that no marketing activity occured for those specific dates and channles.

Outliers

The inspection of potential outliers in the purchase metrics, related to volume and revenue, and media spend variables reveals that extreme observations appear to be consistent with business fluctuations rather than errors or anomalies. Thus, **no values are removed or adjusted**.



Purchasing Trend

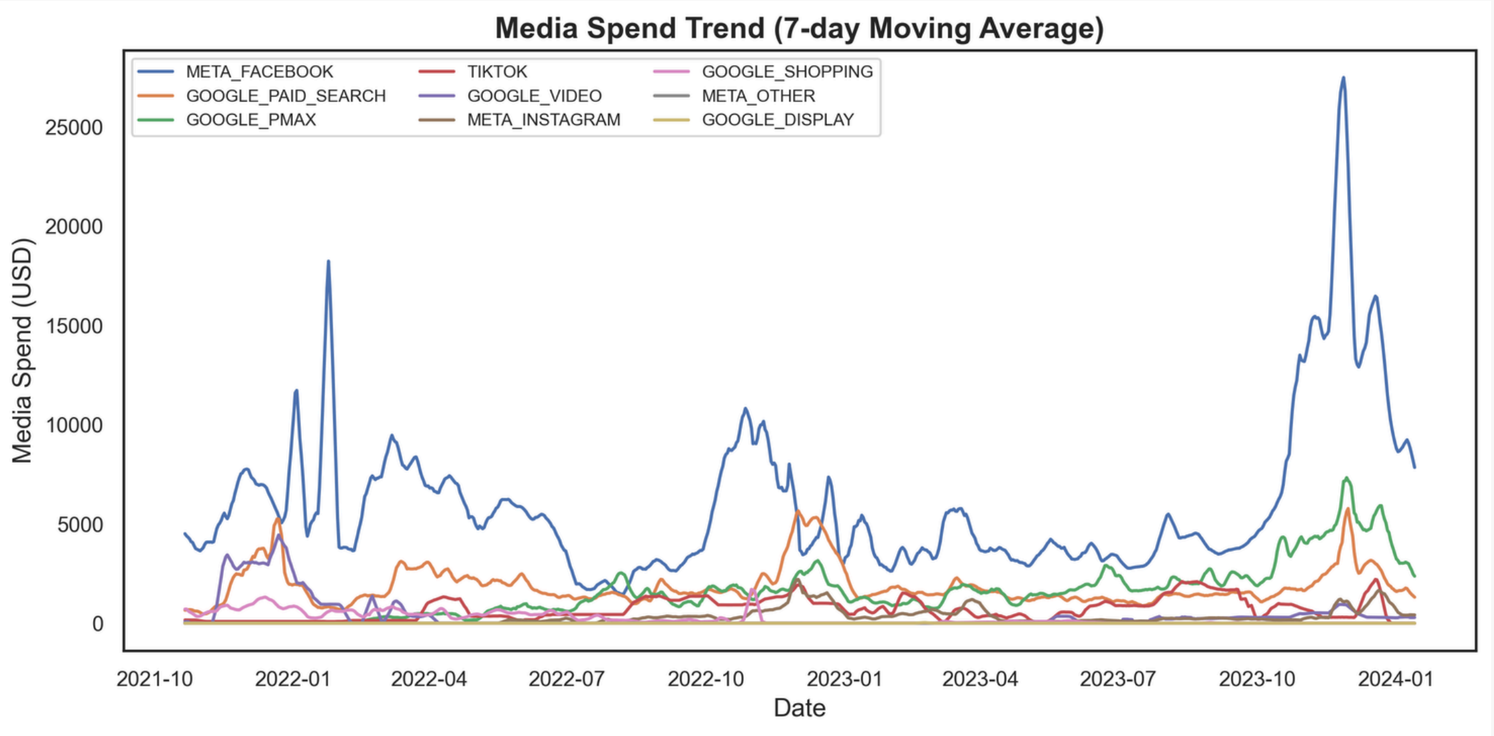
The dataset covers a period of **821 days**, from 15 October 2021 to 13 January 2024, and this plot displays a **7-day moving average** to filter out daily noise and show the trends of both revenue and order volume. As expected, the two lines follow a similar path, confirming a direct relationship between the number of orders and total revenue. Moreover, while the daily averages are about $ 36,990 in revenue and 362 orders, the organisation experienced clear seasonal patters. Indeed, there were recurring increases in the final months of the year, which is common in the Food & Drink sector, with daily revenue reaching $ 120,000 across 1,000 orders by the end of 2023.

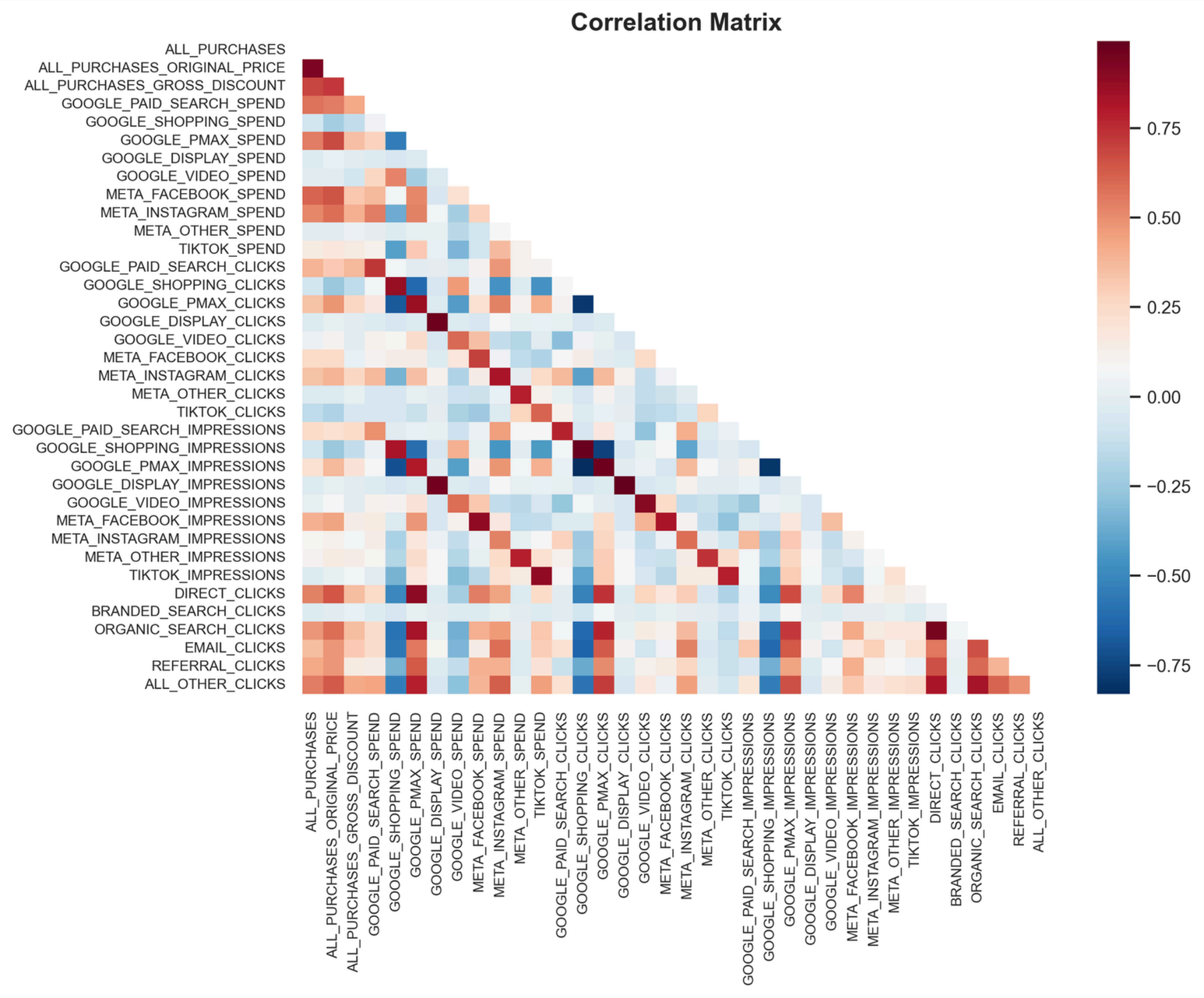
Media Spend Trend

The organisation allocates its **marketing budget across 9 different channels**, which vary in both coverage and total investment. In particular, Facebook is the main media channel, accounting for approximately $ 4.9 million, over 54% of the total spend, with active campaigns throughout the entire period. Then, Google Paid Search is the second largest channel, representing about 17% of the total investment with $ 1.5 million.

The 7-day moving average trend shows that media spending fluctuated in line with the revenue and order patterns. Furthermore, investments consistently increased during the final months of each year, confirming that the business activity is related to seasonal demand.

Spending Channel	Active Days	Percentage Coverage	Total Spending	Percentage Total Spending
GOOGLE_PAID_SEARCH	821	100	1,539,227.37	17.02
GOOGLE_SHOPPING	525	63.95	193,037.70	2.13
GOOGLE_PMAX	703	85.63	1,277,452.99	14.12
GOOGLE_DISPLAY	9	1.1	226.13	0
GOOGLE_VIDEO	382	46.53	311,715.00	3.45
META_FACEBOOK	821	100	4,913,259.52	54.32
META_INSTAGRAM	591	71.99	244,363.48	2.7
META_OTHER	590	71.86	1,199.45	0.01
TIKTOK	798	97.2	564,902.07	6.25





Correlation Analysis

The correlation matrix shows how different variables interact each other. As expected, there is a **strong positive correlation between spending, clicks, and impressions associated with each media channel**, suggesting that marketing budget allocation attract users across all platforms. Moreover, the plot displays also a **positive correlation between overall marketing spend and purchase metrics**, related to revenue and order volumes, highlighting the impact of media investment toward sales. In contrast, non-paid media channels appear isolated from other marketing activities. Indeed, while these channels are strongly correlated with one another, they do not show any meaningful connection to daily changes in advertising spend.

METHODOLOGY

MODELLING APPROACH AND MERIDIAN CONFIGURATION

Feature Engineering

After verifying the daily aggregation and the absence of missing values, the feature set for the Google Meridian MMM was organized as shown in the table on the right.

In particular, channels with negligible spend (GOOGLE_DISPLAY and META_OTHER) were excluded, as well as the organic media BRANDED_SEARCH_CLICKS, due to a 98% missing data rate. Moreover, the feature ALL_PURCHASES_GROSS_DISCOUNT accounts for discount rates only for completed purchases, so it could introduce data leakage if considered as a control variable.

Holdout Strategy

Following Meridian recommendations, a **distributed holdout approach** was implemented for performing the train-validation split by **selecting one day out of every four** (25% of the available data) across the entire time series. This solution ensures that both the training and validation splits are representative of the full period, allowing for a reliable estimation of the model parameters and the evaluation metrics.

Category	Description	Features
Target	Revenue	ALL_PURCHASES_ORIGINAL_PRICE
Media Spend	Marketing investment across paid channels	GOOGLE_PAID_SEARCH_SPEND, GOOGLE_SHOPPING_SPEND, GOOGLE_PMAX_SPEND, GOOGLE_VIDEO_SPEND, META_FACEBOOK_SPEND, META_INSTAGRAM_SPEND, TIKTOK_SPEND
Media Exposure	Main engagement metrics associated with each spend channel	GOOGLE_PAID_SEARCH_CLICKS, GOOGLE_SHOPPING_CLICKS, GOOGLE_PMAX_CLICKS, GOOGLE_VIDEO_IMPRESSIONS, META_FACEBOOK_IMPRESSIONS, META_INSTAGRAM_IMPRESSIONS, TIKTOK_IMPRESSIONS
Control Variables	Organic traffic sources without spend	DIRECT_CLICKS, ORGANIC_SEARCH_CLICKS, EMAIL_CLICKS, REFERRAL_CLICKS, ALL_OTHER_CLICKS
	Temporal features	DAY_OF_WEEK, IS_WEEKEND, MONTH

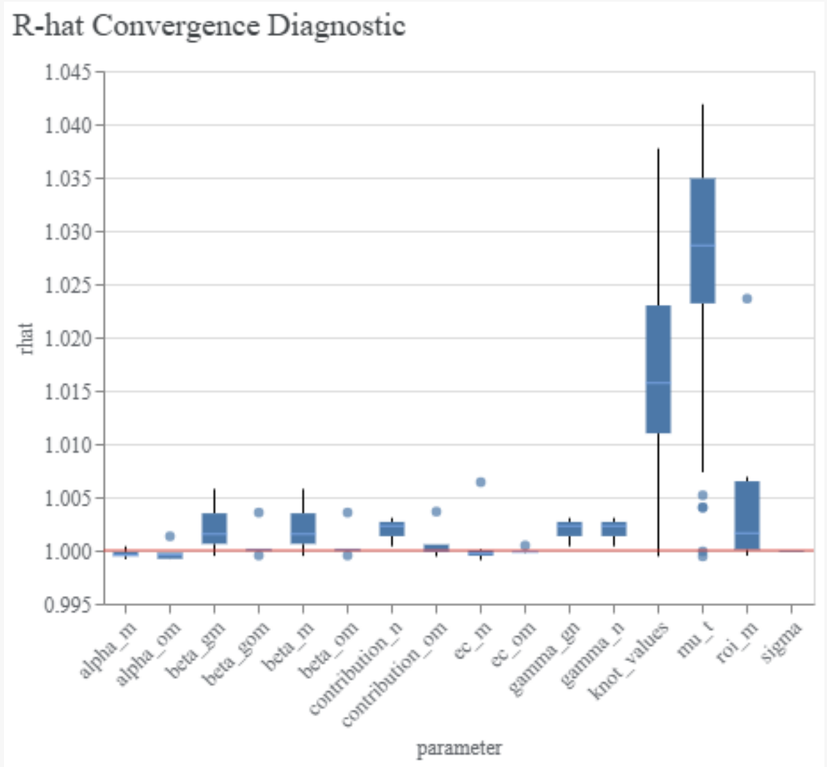
Model Configuration

After the preprocessing step, the Meridian framework requires to map each variable to its role, distinguishing between paid media, organic media, and non-media treatments, together with the integration of the holdout index. Regarding the model specification, **LogNormal priors** were defined for each channel ROI, as they ensures positive estimates and models the typical right-skewness of marketing performance. In this context, specific location and scale parameters were applied by assigning higher expectations to search and shopping channels, while adopting broader distributions for video and social platforms. Moreover, the **knot scaling** was enabled to ensure the model is able to automatically capture the temporal variations in media effectiveness. Ultimately, the model was trained using a **Bayesian MCMC approac**h by performing both prior sampling to validate the model assumptions and posterior sampling to extract reliable parameter estimates.

Model Convergence

The trained model passed all **quality and diagnostic checks**. In particular:

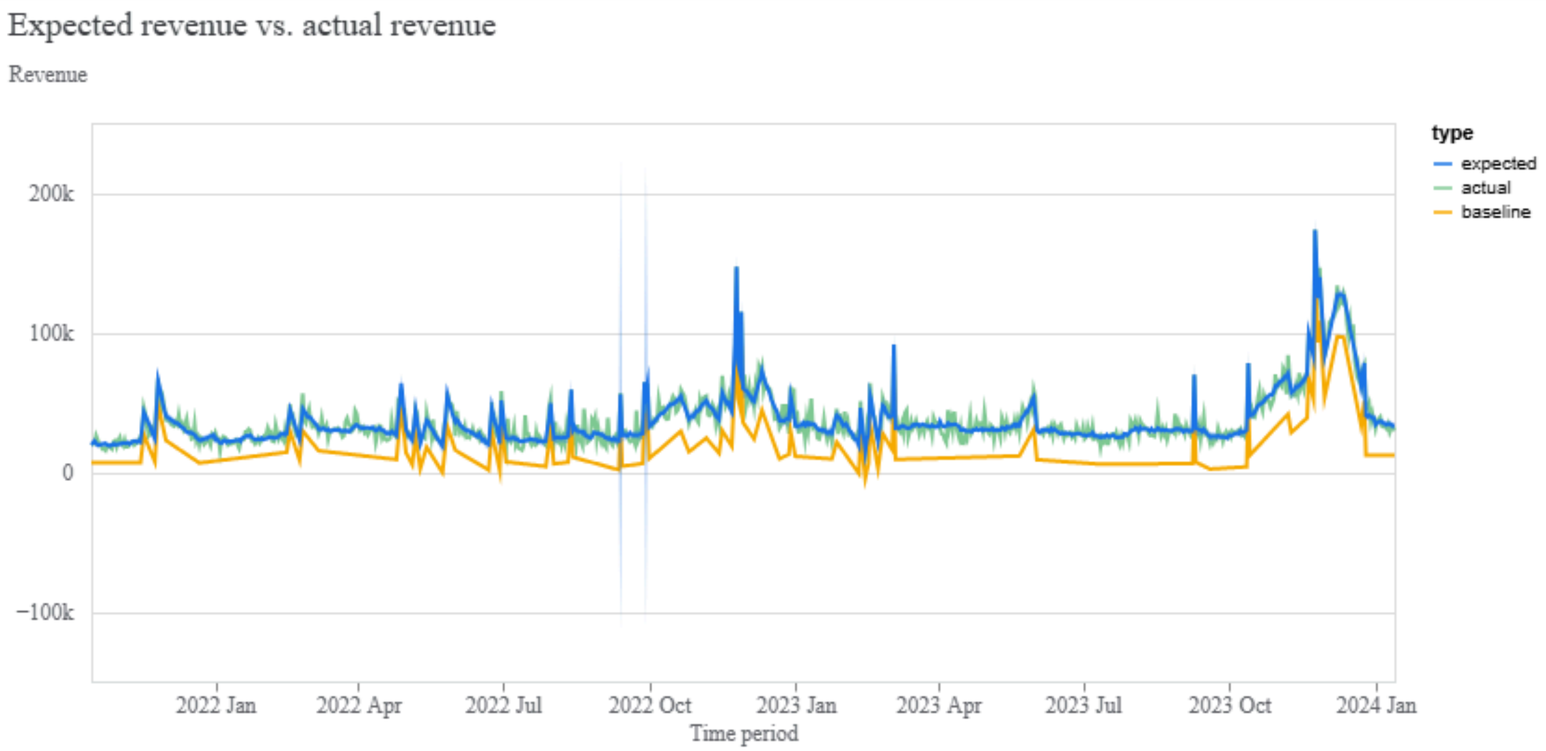
- The model converges, as all parameters have R-hat values < 1.2.
- The shift from prior to posterior distributions indicates that the model has effectively learned from the data.
- The ROI estimates are within a reasonable range, suggesting that the prior assumptions provide a reliable foundation for the model.

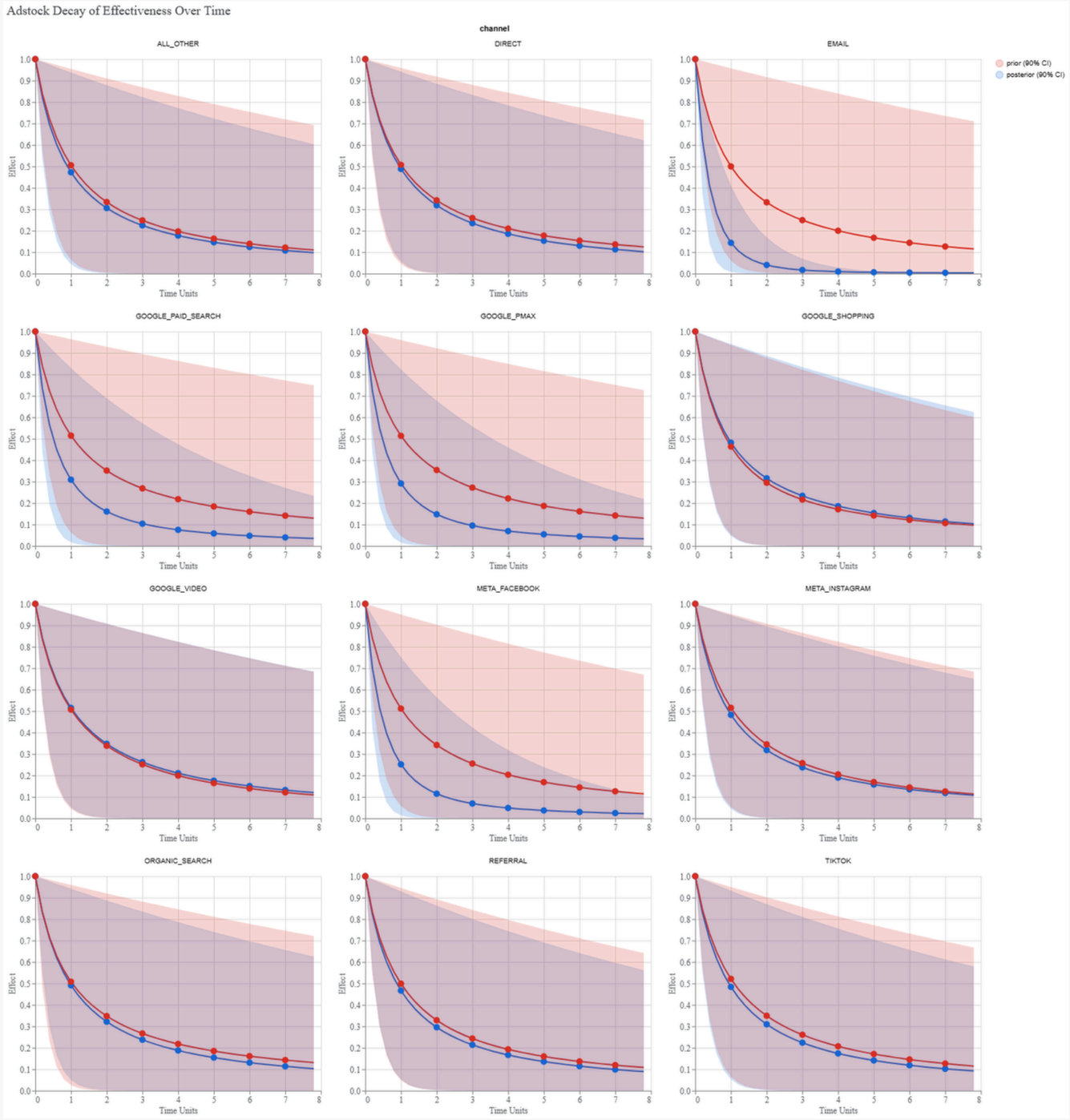


Model Quality and Metrics

The trained model demonstrates a high level of **predictive accuracy and reliability**. In particular, it shows similar performance across both training and test sets, preventing overfitting and explaining over 94% of the total revenue variance. Moreover, the Bayesian posterior predictive p-value of 0.74 confirms that the observed data is consistent with the model's predictions. Considering the plot, the blue line of expected revenue closely follows the green one of actual revenue throughout the considered period, with the baseline orange line allowing the distinction of the incremental lift generated by media investments.

Metric	Evaluation Set	Value
R_Squared	Train	0.9501
R_Squared	Test	0.9131
R_Squared	All Data	0.9416
MAPE	Train	0.1061
MAPE	Test	0.1282
MAPE	All Data	0.1116
wMAPE	Train	0.0933
wMAPE	Test	0.1167
wMAPE	All Data	0.0991

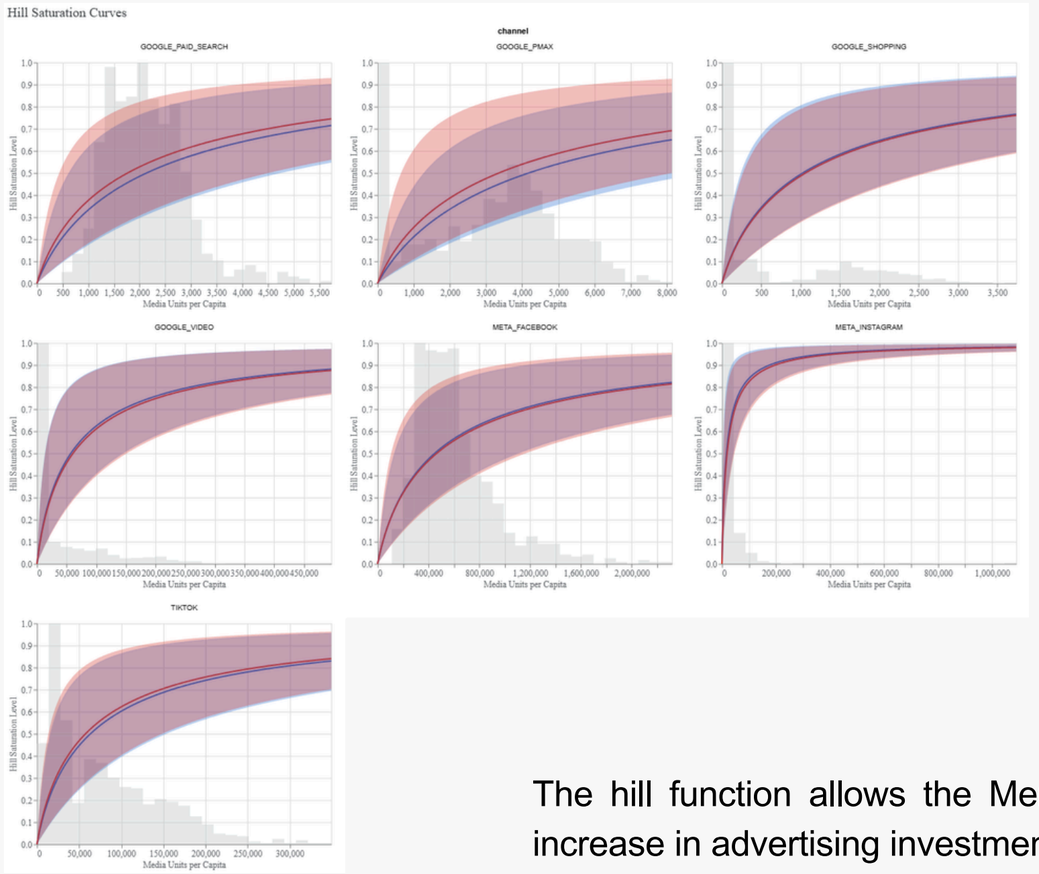


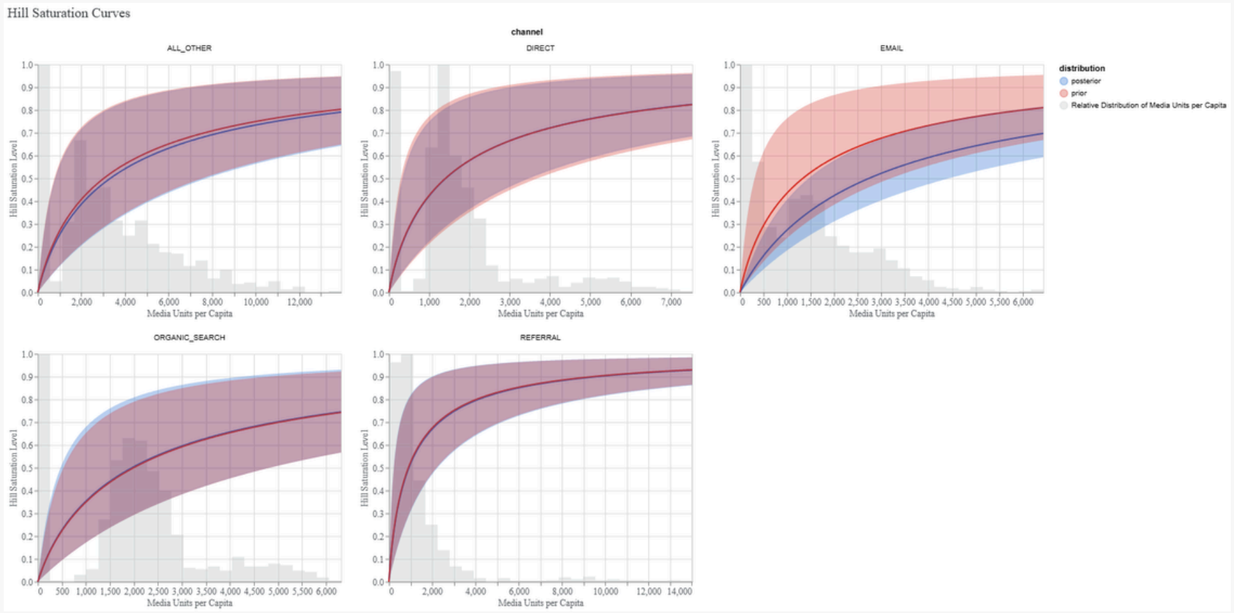


Adstock Effects

The adstock function allows the Meridian model to capture the **carryover effect**, assuming that advertising does not affect sales only in the current period, thereby better understanding customer purchasing behaviour.

There are some differences in how specific media channels influence revenue over time. In particular, Google Shopping and Meta Facebook display a faster decay, suggesting that they are primary drivers for immediate conversions. Instead, Google Paid Search and Google PMax show a slower decay, indicating that advertising exposure continues to influence sales over time.





Hill Saturation Effects

The hill function allows the Meridian model to capture the **saturation effect**, assuming that an increase in advertising investments does not produce proportionally increasing returns indefinitely.

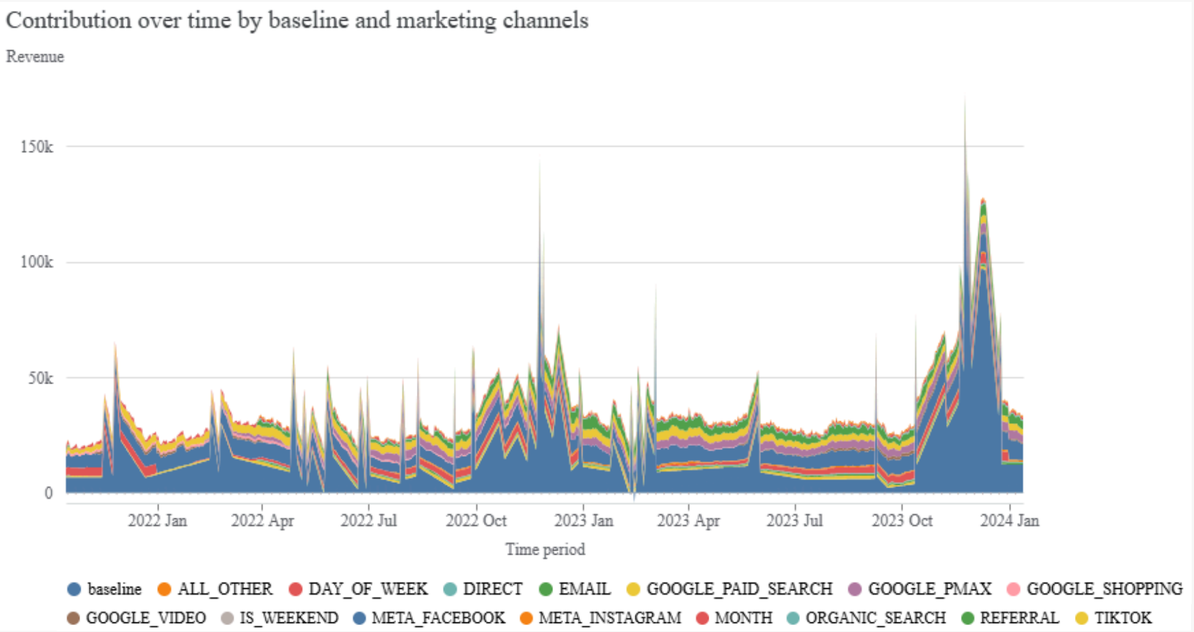
Channel efficiency varies significantly according to how quickly each media platform reaches saturation. In particular, Meta Instagram shows a clear initial increase in revenue even with limited marketing budget, but it also reaches diminishing returns sooner. Instead, other channels like Google Paid Search and Google PMax display a more gradual growth curve, suggesting that they can handle larger investments over time without losing efficiency as quickly.

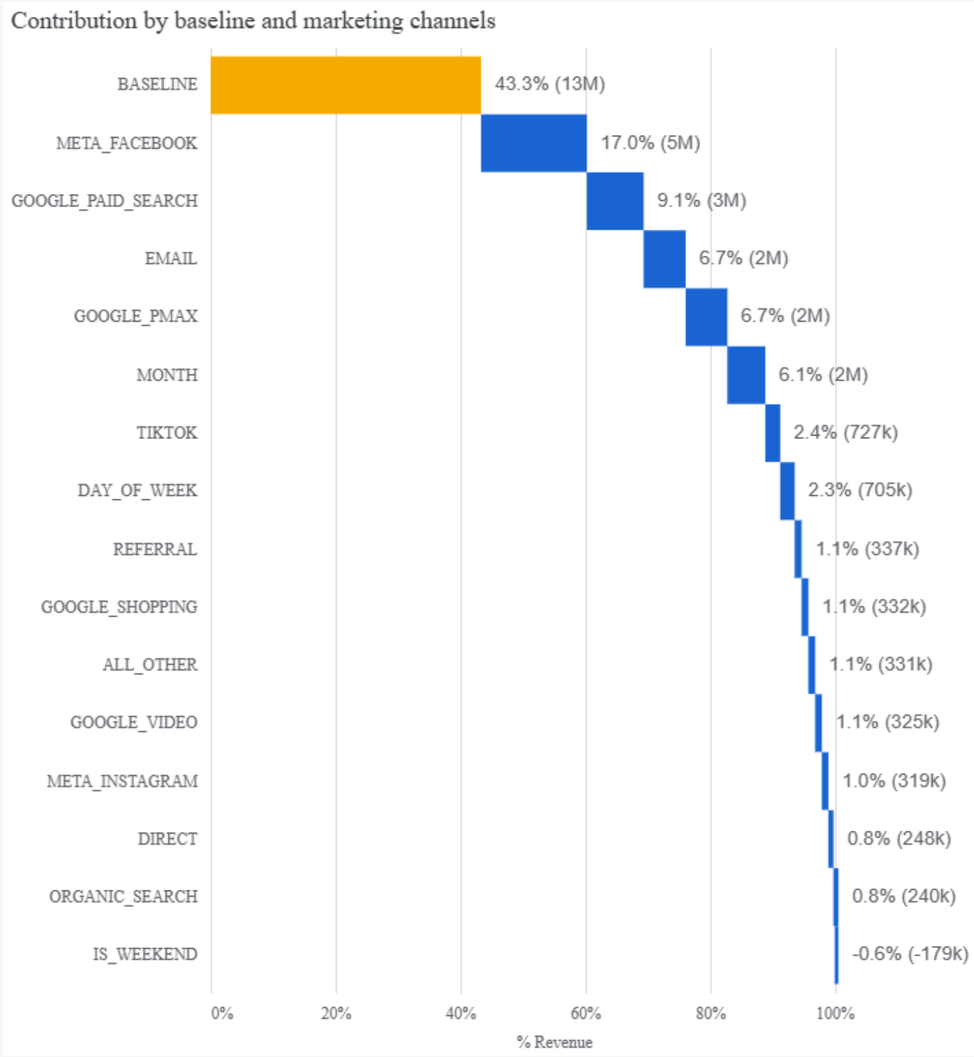
RESULTS

CHANNEL CONTRIBUTIONS

Features Contributions and Trend

Considering the waterfall chart, the **baseline component** accounts for 43.3% of total revenue, representing the expected volume without active marketing. Among paid channels, **Meta Facebook** is the primary driver at 17%, followed by **Google Paid Search**. Together, they represent the most significant advertising investments. Ultimately, the area chart shows the stability of the baseline level over time, while also highlighting the fundamental contribution of media channels to the total revenue share.





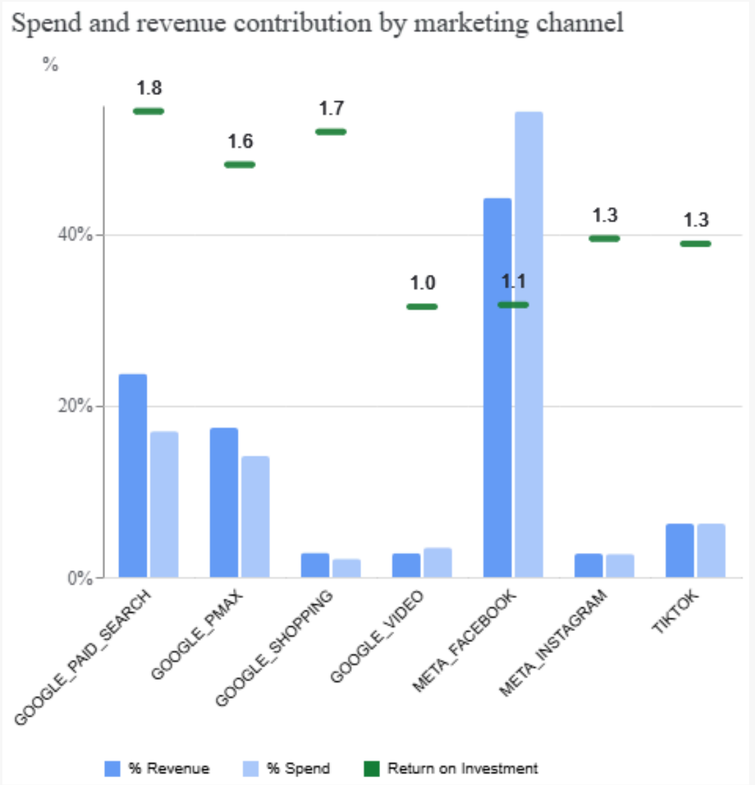
RESULTS

ROI AND MARGINAL ROI

Incremental Contribution by Channel

Combining channel contribution with ROI analysis highlights that Meta Facebook is the main driver for incremental sales, accounting for 44.25% of the total, but it has a limited ROI of 1.05. On the other hand, Google Paid Search is the second largest media by volume and provides the best ROI at 1.80. Google Shopping and PMax also show strong efficiency, with ROIs of 1.72 and 1.59, respectively.

Channel	Incremental Outcome	Contribution Percentage	Cumulative Percentage
GOOGLE_PAID_SEARCH	2,767,801.50	23.72	23.72
GOOGLE_SHOPPING	331,778.78	2.84	26.56
GOOGLE_PMAX	2,033,691.50	17.43	43.99
GOOGLE_VIDEO	325,342.50	2.79	46.78
META_FACEBOOK	5,163,265.00	44.25	91.03
META_INSTAGRAM	319,298.06	2.74	93.77
TIKTOK	726,789.81	6.23	100
All Channels	11,667,959.00	100	100

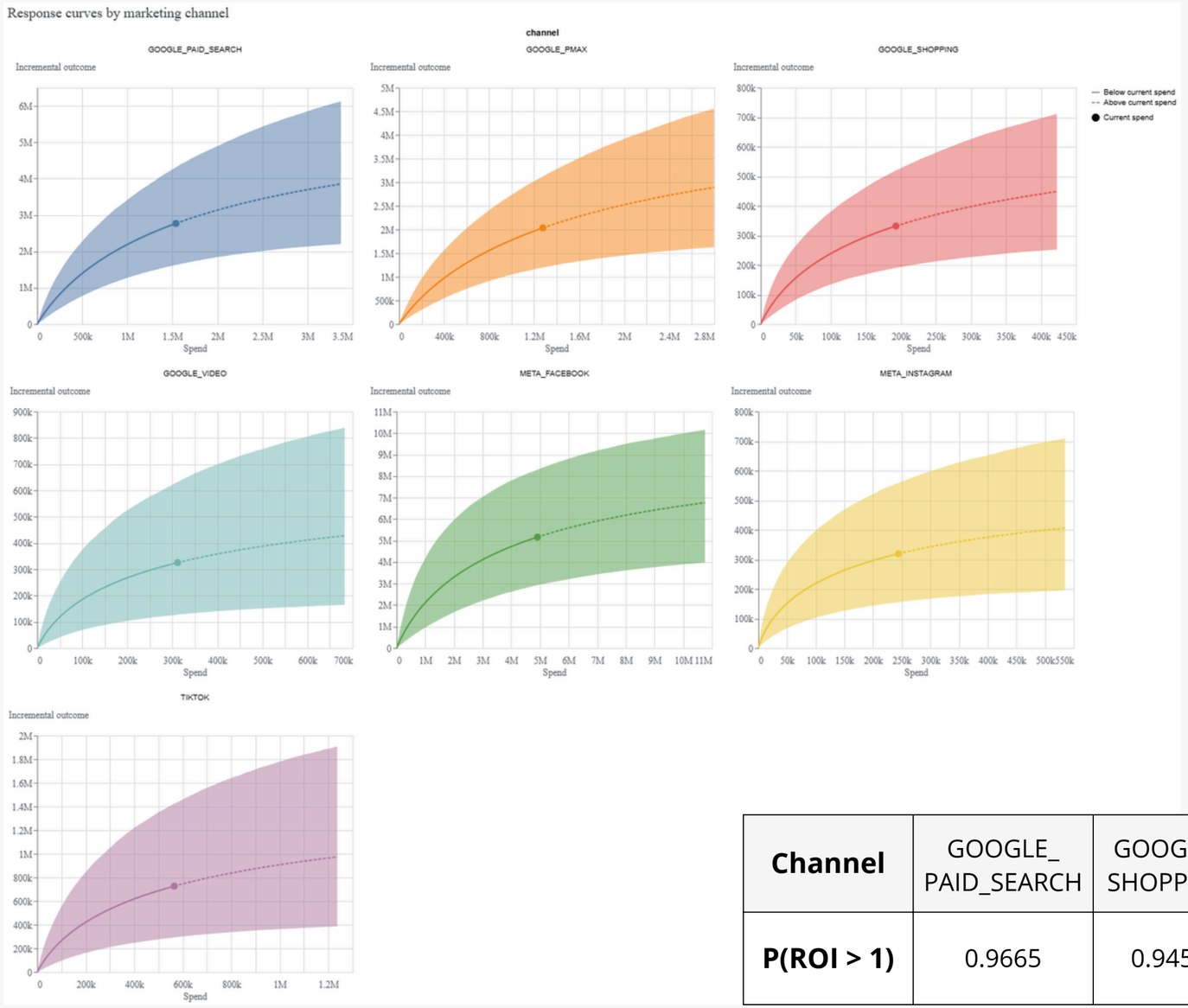


ROI and mROI Tables

Considering both **ROI and marginal ROI for each marketing channel**, along with their corresponding **credible intervals**, Google Paid Search emerges as the most efficient platform, with an ROI of 1.80 and an mROI of 0.90, followed by other media like Google Shopping and Google PMax. In contrast, Meta Facebook is the main driver of revenue volumes, but it shows less efficiency with an ROI of 1.05 and an mROI of 0.43. Therefore, while **Meta is essential for reaching large volumes, Google provides the best return for future investments**.

Channel	ROI mean	ROI median	ROI CI lower	ROI CI Upper
GOOGLE_ PAID_ SEARCH	1.7982	1.7194	1.0572	2.7996
GOOGLE_ SHOPPING	1.7187	1.6416	0.9873	2.6928
GOOGLE_ PMAX	1.592	1.5349	0.9374	2.4443
GOOGLE_ VIDEO	1.0437	0.9427	0.4093	2.0273
META_ FACEBOOK	1.0509	0.9995	0.5966	1.6886
META_ INSTAGRAM	1.3067	1.2098	0.6367	2.2846
TIKTOK	1.2866	1.1611	0.5193	2.5196
All Channels	1.2901	1.268	0.9817	1.6858

Channel	mROI mean	mROI median	mROI CI lower	mROI CI Upper
GOOGLE_ PAID_ SEARCH	0.896	0.8538	0.3561	1.5666
GOOGLE_ SHOPPING	0.7612	0.7177	0.3324	1.3228
GOOGLE_ PMAX	0.8345	0.799	0.3572	1.4254
GOOGLE_ VIDEO	0.4219	0.3679	0.1331	0.8827
META_ FACEBOOK	0.4334	0.4293	0.2027	0.6752
META_ INSTAGRAM	0.4686	0.4301	0.1706	0.8923
TIKTOK	0.5591	0.4921	0.1738	1.1652
All Channels	NaN	NaN	NaN	NaN



Channel Efficiency

Analysing the probability of achieving a ROI greater than 1, it is possible to identify **which channels are potentially over-invested or under-invested**.

In particular, Google channels, including Paid Search, Shopping, and PMax, show high probabilities, ranging from 92% to 97%, of delivering a positive return. Looking at their **response curves**, these channels are currently performing well despite being under-invested. Instead, Meta Facebook and Google Video show lower probabilities, below 50%, of achieving a ROI greater than 1. Based on their current spend levels and the diminishing returns shown in their response curves, the company may be over-investing in these areas.

Channel	GOOGLE_ PAID_SEARCH	GOOGLE_ SHOPPING	GOOGLE_ PMAX	GOOGLE_ VIDEO	META_ FACEBOOK	META_ INSTAGRAM	TIKTOK
P(ROI > 1)	0.9665	0.9454	0.9196	0.4525	0.4998	0.6894	0.6213

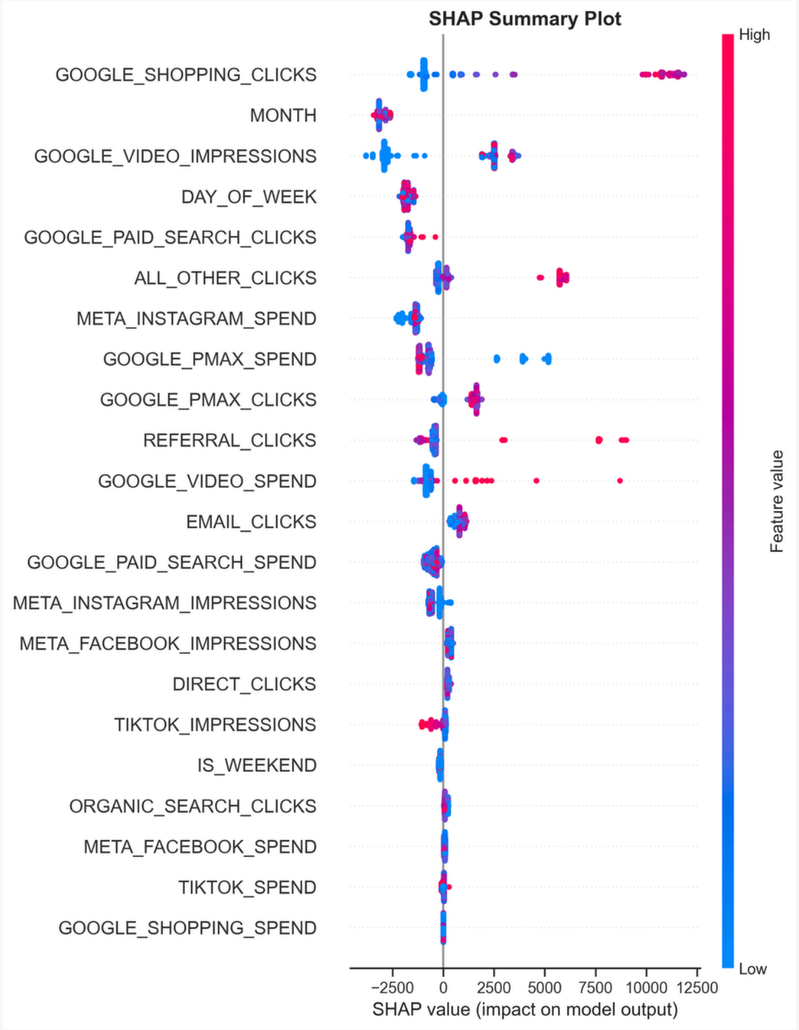
SHAP EXPLAINABILITY

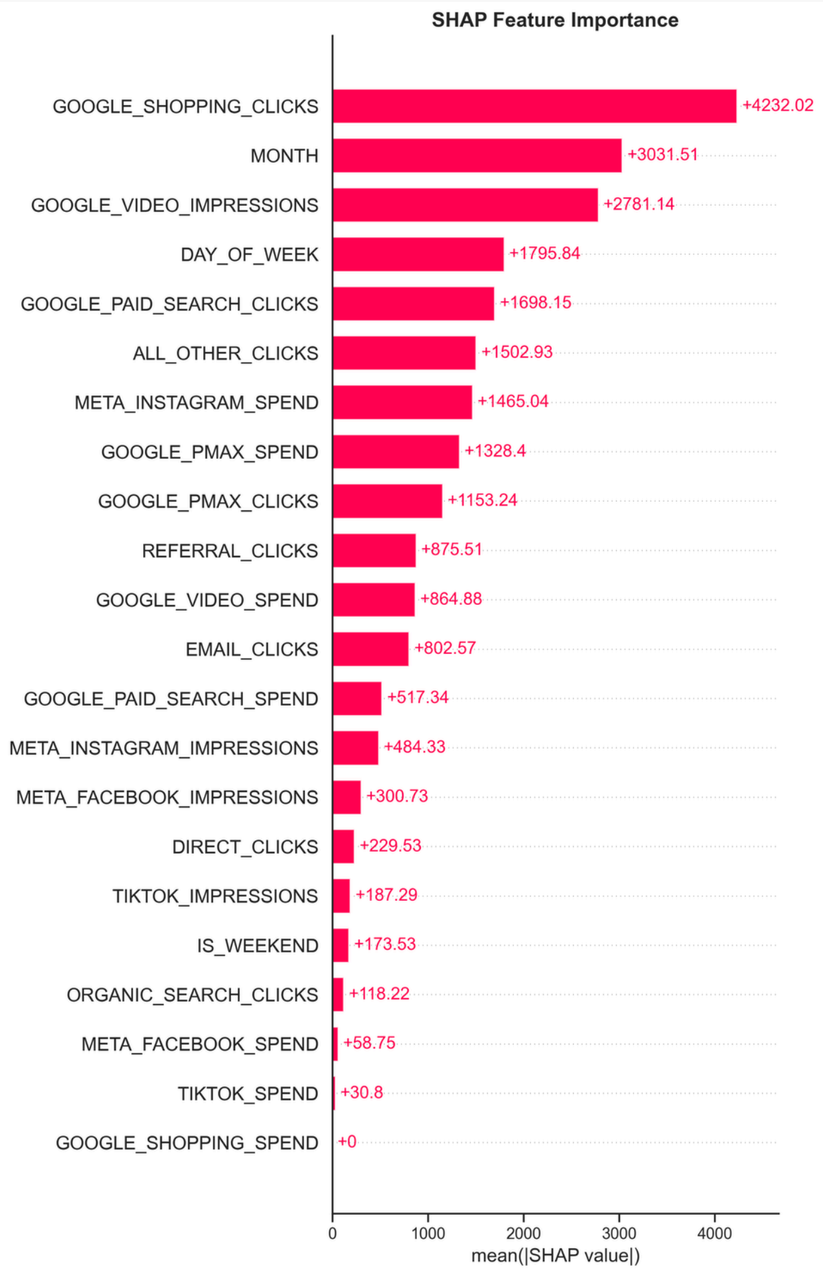
Surrogate Model

To better understand the actual drivers behind MMM performance, Shapley values were used to interpret the model's results. In particular, since Google Meridian is a Bayesian framework, it is not possible to apply SHAP directly. Therefore, a surrogate model was built using a **HistGradientBoostingRegressor**, which was trained using the same feature set and the same holdout dataset as the Meridian model. The ML model performed well, achieving an R-squared of 0.8242 on the test set, being able to capture the underlying patterns and generalizes well to unseen data.

Tree Explainer and Shapley Values

Using the TreeExplainer on the surrogate model, it is possible to extract insights on how each marketing channel contributes to the overall revenue. In this context, the **summary plot** displays how each feature affects the outcome. In particular, the variable GOOGLE_SHOPPING_CLICKS has a strong positive impact on the model's predictions when its values are high, whereas when the values of this feature are low, the **Shapley values** shift to the negative side. A similar but lighter behaviour applies to other variables, such as REFERRAL_CLICKS and GOOGLE_VIDEO_SPEND, further reinforcing the importance of maintaining marketing activity on these channels. Beyond the main drivers, the summary plot also shows other variables with a marginal impact on the model, thereby obtaining Shapley values highly concentrated around the centre and making their individual contribution harder to distinguish or interpret.





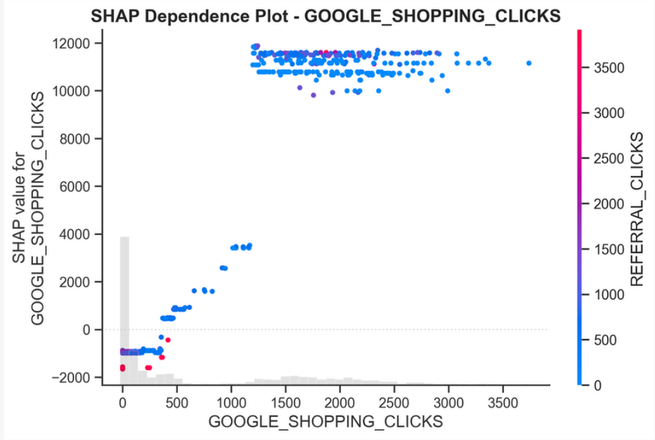
Feature Importance and Comparison

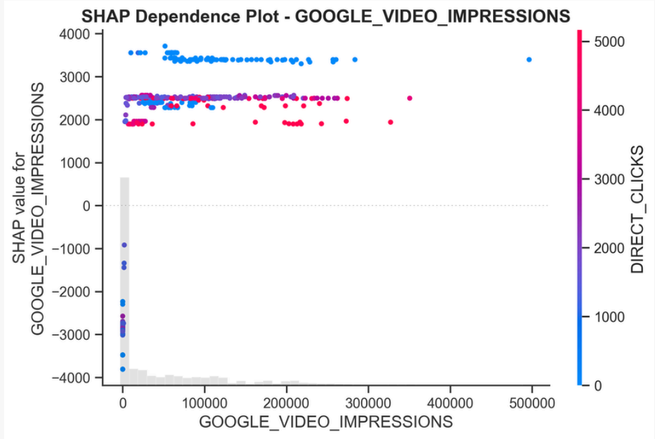
The **feature importance plot** highlights that GOOGLE_SHOPPING_CLICKS is the most significant feature, followed by MONTH and GOOGLE_VIDEO_IMPRESSIONS, suggesting that Google performance is essential for the company. Temporal variables also rank high, as seasonality and recurring patterns play an important role in affecting revenue. Moreover, it is possible to **compare SHAP importance from the surrogate model with the actual contribution estimates from the Meridian MMM** in order to better understand the performance of media channels. In particular, the SHAP analysis is focused on short-term fluctuations, highlighting Google channels as primary drivers of daily revenue, while the Meridian model better captures the long-term effects of channels like Meta Facebook, which represents the largest investment over the considered period.

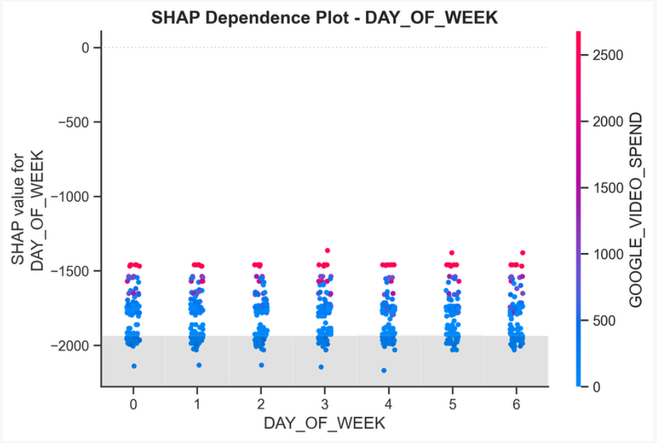
Channel	SHAP Importance	Importance Percentage	MMM Contribution	Contribution Percentage
GOOGLE_PAID_SEARCH	2,215.49	14.67	2,767,801.50	23.72
GOOGLE_SHOPPING	4,232.02	28.02	331,778.78	2.84
GOOGLE_PMAX	2,481.65	16.43	2,033,691.50	17.43
GOOGLE_VIDEO	3,646.02	24.14	325,342.50	2.79
META_FACEBOOK	359.48	2.38	5,163,265.00	44.25
META_INSTAGRAM	1,949.38	12.91	319,298.06	2.74
TIKTOK	218.09	1.44	726,789.81	6.23

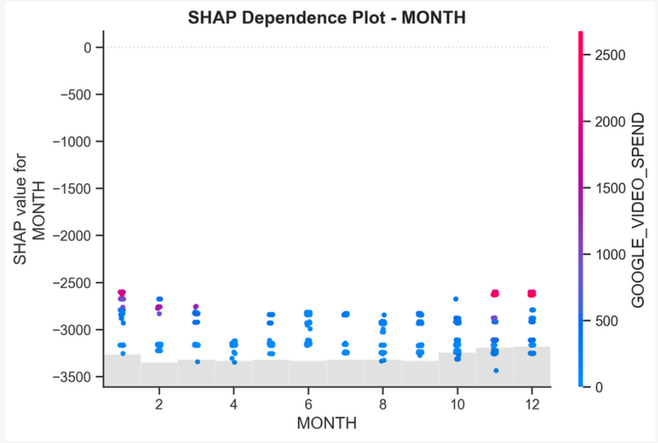
Analysis of Key Drivers

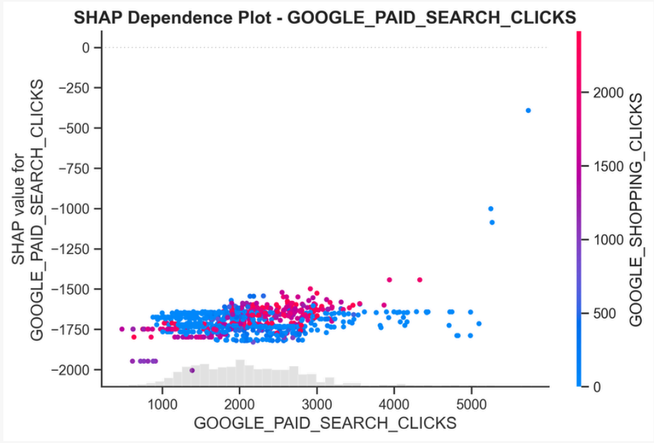
To further inspect the characteristics of the top-5 drivers, the **dependence plots** show how each variable affects the model's predictions. It is important to recall that the color intensity of the data points refers to the influence of a secondary interacting variable. Considering the feature GOOGLE_SHOPPING_CLICKS, a pattern of diminishing returns is observed. Indeed, after reaching a certain volume of clicks, the additional impact on the model becomes lighter. This effect is non-linear, shifting from a negative impact around zero to a strong positive influence as click volume increases. Moreover, regarding GOOGLE_VIDEO_IMPRESSIONS, when the values are near zero, the model shows a negative impact, but once the company achieves a minimum level of visibility, the contribution becomes consistently positive. At the same time, GOOGLE_PAID_SEARCH_CLICKS displays an approximately linear trend, as increasing these clicks provides gradual benefits. Ultimately, the categorical variables MONTH and DAY_OF_WEEK capture seasonality, since specific time periods have a great impact on the company's results.











SCENARIO PLANNING
INVESTMENT SIMULATIONS

Scenario Simulation General Setting

The current marketing investment framework is based on a total expenditure of approximately $ 9.04 million. By analysing the efficiency of various media channels in terms of ROI and marginal ROI, **Google Paid Search acts as the most performing platform for generating revenue**. In constrast, Google Video, Meta Facebook, and Meta Instagram reach a state of saturation, suggesting that additional investments may lead to lower incremental returns compared to other channels.

Scenario Types

To perform an efficient reallocation of current and future marketing budget, **four different investment scenarios are analysed under various strategies**. The first scenario focuses on the impact of a 20% increase in total budget, while the second scenario doubles the investment for the most performing channel. Moreover, the third scenario assesses the gains from reducing spending on saturated channels by 30%, whereas the fourth scenario evaluates the effects of a 10% budget increase related to a random half of the channels. These scenarios are compared by considering **three allocation strategies: equal allocation, mROI allocation, and saturated channels allocation**.

Scenario	Budget	Incremental Revenue	Uplift	ROI
Scenario 1 - Equal Allocation	10,852,749.75	11,577,234.00	1,038,910.00	1.067
Scenario 1 - mROI Allocation	10,852,749.75	11,545,538.00	858,234.00	1.0631
Scenario 1 - Saturated Channels Allocation	10,852,749.75	12,945,485.00	85,873.00	1.1931
Scenario 2 - Equal Allocation	9,813,571.81	11,063,965.00	1,007,880.00	1.1274
Scenario 2 - mROI Allocation	9,813,571.81	11,058,465.00	842,233.00	1.1269
Scenario 2 - Saturated Channels Allocation	9,813,571.81	12,409,667.00	88,557.00	1.2646
Scenario 3 - Equal Allocation	7,403,156.73	9,677,276.00	926,992.00	1.3067
Scenario 3 - mROI Allocation	7,403,156.73	9,728,026.00	779,867.00	1.3142
Scenario 3 - Saturated Channels Allocation	7,403,156.73	10,910,977.00	93,650.00	1.4741
Scenario 4 - Equal Allocation	9,708,510.59	11,009,082.00	1,004,239.00	1.1339
Scenario 4 - mROI Allocation	9,708,510.59	11,007,695.00	840,442.00	1.1339
Scenario 4 - Saturated Channels Allocation	9,708,510.59	12,351,705.00	88,136.00	1.2725

SCENARIO PLANNING

BUSINESS RECOMMENDATIONS

Business Insights and Budget Optimization Recommendations

The overall analysis highlights Meta Facebook as the primary driver of incremental revenue, accounting for over 44% of total channel contribution. However, this media channel also exhibits one of the lowest marginal ROIs, suggesting that it is almost saturated. Instead, Google Paid Search, Google Shopping, and Google PMax deliver a ROI greater than 1 and represent the most efficient platforms for future investments. Moreover, the saturation analysis shows that Google Video, Meta Facebook, and Meta Instagram produce diminishing returns, meaning that increasing the budget in these areas will not lead to much more growth.

To maximize future revenue and optimize budget allocation for the next quarter, instead of simply increasing total spend, the most effective strategy prioritises advertising investment in those channels that have not yet reached their saturation point. Indeed, the investment simulation across four different scenarios reveals that **an allocation strategy that manages the budget in under-performing and saturated platforms, by redistributing it across the top performing channels, yields the highest return on investment**.

Ultimately, **the best approach toward revenue maximization is to focus on the efficiency of marketing investments** rather than on the overall volume. In particular, staying active on social platforms is essential for brand awareness and perception, but further spending should address direct search channels where customers play an active role.

LIMITATIONS

Data Quality

The advertising spend and revenue data are already aggregated at a daily level, requiring no additional transformation. Moreover, some missing values are included for media channel variables, such as spend, clicks, and impressions. However, they are simply imputed with a value of zero, assuming that they represent a total absence of advertising activity rather than a data collection error.

Causal Assumptions

Google Meridian relies on a Bayesian framework for causal inference. The main assumption is called **conditional exchangeability**, meaning that after accounting for observed covariates, including media activities and seasonality, the advertising spend is independent of any unobserved confounders that may also influence overall revenue. Additionally, the integration of prior distributions based on domain knowledge typically constrains the model's parameters to realistic ranges, drawing reliable conclusions.

Model Uncertainty

First, **credible intervals** are used to quantify the range of plausible values for ROI and mROI estimates, as Meridian adopts a Bayesian probabilistic approach. Then, the definition of a **surrogate model** to better interpret the MMM through SHAP analysis may introduce slight discrepancies due to the different modelling approaches underlying the two models.